The examination is being carried out on the **following application documents**

Description, Pages

1-25 as originally filed

Claims, Numbers

1-15 as originally filed

Drawings, Sheets

1/7-7/7 as originally filed

1 Prior art

Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 CN 112 258 898 A (CAAC EASTERN CHINA REGIONAL CENTRAL TRAFFIC ADMINISTRATION) 22 January 2021

D2 EP 3 534 351 A1 (HONEYWELL INT INC [US]) 4 September 2019

2 Articles 54 and 56 EPC

2.1 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claims 1-6, 12-15 is not new within the meaning of Article 54(1) and (2) EPC, the present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claims 7-11 does not involve an inventive step within the meaning of Article 56 EPC.

2.2 *Independent claim 1*

2.2.1 Document D1 discloses a computerized method for controlling an aerial vehicle to travel in an air corridor while maintaining safe separation from other agent vehicles (D1: paragraph 6), the method comprising:

A. receiving, by a computer processor of an aerial vehicle, a trained air corridor model associated with an origin, a destination, and an air corridor representative of airspace between the origin and the destination (D1: paragraphs 6, 11, 39, 68; claim 1)

B. wherein the trained air corridor model is configured to maintain safe separation distance between vehicles in the air corridor (D1: paragraphs 59, 72; claim 6)

C. identifying, by the computer processor, current position data representing a current position of the aerial vehicle in the air corridor (D1: paragraphs 30, 31; claims 1, 2, 7)

D. receiving, by the computer processor, agent position data associated with a position of at least one agent vehicle in the air corridor (D1: paragraphs 30, 31; claims 1, 2, 7)

E. and determining, by the computer processor, a next position of the aerial vehicle in the air corridor based on the trained air corridor model, the current position data, and the agent position data (D1: claims 1, 2).

The subject-matter of claim 1 is therefore not new over D1 (Article 54(1) and (2) EPC).

2.2.2 Document D2 discloses features A-E (D2: paragraphs 2, 4, 12, 14, 16, 17). The subject-matter of claim 1 is therefore not new over D2 (Article 54(1) and (2) EPC).

2.3 *Independent claims 12, 14, 15*

The same reasoning as for claim 1 can be applied, *mutatis mutandis*, to independent claims 12, 14, 15. The subject-matter of claims 12, 14, and 15 is therefore not new over D1 or D2 (Article 54(1) and (2) EPC).

2.4 *Dependent claims 2-11, 13*

2.4.1 Dependent claims 2-11, 13 do not appear to contain any additional features which, in combination with the features of any claim to which they refer, meet the requirements of the EPC with respect to novelty and/or inventive step.

I. Claim 2: the division of the airspace into data cubes is anticipated by D1 (D1: paragraphs 58-60) and D2 (D2: paragraph 16; claim 1), not novel.

II. Claim 3: the size of the cube being based on minimum safe distance is anticipated by D1 (D1: paragraph 60), not novel.

III. Claim 4: the identification of the cube wherein the aircraft is located is anticipated by D1 (D1: paragraph 64) and D2 (D2: claim 1), not novel.

IV. Claim 5: anticipated by D1 (D1: paragraphs 58-60) and D2 (D2: paragraphs 14, 16; claim 1), not novel.

V. Claim 6: optimizing airspace is anticipated by D1 (D1: paragraph 39) and D2 (D2: paragraph 17), not novel.

VI. Claims 7, 8: the division or combination of cubes into smaller or larger cubes is well known in the field of numerical analysis, and only involves well-known trade-off considerations between computational accuracy and speed. Not inventive.

VII. Claim 9: exchanging position data between aircraft is notoriously well known. Not inventive.

VIII. Claim 10: controlling an aircraft according to air traffic data is common general knowledge, not inventive.

IX. Claim 11: this altitude range is arbitrary and could be covered by the disclosures of D1 and D2 without modifications. Not inventive.

X. Claim 13: the aerial vehicle is anticipated by D1 and D2, not novel.

3 Further remarks

3.1 The independent claims are not in the two-part form in accordance with Rule 43(1) EPC, which in the present case would be appropriate, with those features known in combination from the prior art being placed in the preamble (Rule 43(1)(a) EPC) and the remaining features being included in the characterising part (Rule 43(1)(b) EPC).

3.2 The features of the claims should be provided with reference signs placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC). This applies to both the preamble and characterising portion (see Guidelines F-IV, 4.19).

3.3 To meet the requirements of Rule 42(1)(b) EPC, D1 and D2 should be identified in the description and the relevant background art disclosed therein should be briefly discussed.